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Tensor CSAMT and AMT Studies of the Xiarihamu Ni-Cu Sulfide Deposit in Qinghai, China

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ABSTRACT

The frequency domain electromagnetic sounding method is one of the important means of exploring mineral resources. This paper combined tensor controlled source audio frequency magnetotelluric (CSAMT) and audio frequency magnetotelluric(AMT) methods to investigate Xiarihamu copper-nickel ore in Qinghai Province. Spliced tensor CSAMT high frequency (200 Hz–8533 Hz) and AMT low frequency (1 Hz–200 Hz) apparent resistivity and impedance phase data, as well as spliced data, were used for a 2D inversion and to generate a 2D geoelectric model. Coupled with the known geological, geophysical, and drilling data, the inversion results were analyzed. The results show that the combined use of tensor CSAMT and AMT can overcome the inherent defects of these methods, such as limited exploration depth and the "dead band" effect. This method produced good results in terms of exploring the underground electrical structure, finding the low resistivity body, and determining the thickness of the overburden, as well as the occurrence and the buried depth of the ore body. These results provide an effective method and exploration strategy to find similar potential deposits in the surrounding areas.

1. Introduction

The Xiarihamu copper-nickel deposit is located in the western part of the eastern Kunlun Mountains in western Qinghai Province; the southern margin of the Qaidam Basin is approximately 180 km northwest of Golmud. This deposit is a large-scale magmatic liquation copper-nickel sulfide deposit that was recently discovered in the East Kunlun orogenic belt. This area is the second largest copper-nickel deposit (Zhang et al., 2016), second only to the Gansu Jinchuan magmatic copper-nickel sulfide deposit, and it has attracted the attention of many geologists. Geophysical methods are an important method of mineral resource exploration (Liu et al., 2013). It is difficult to carry out ground geophysical surveys in this area due to the steep mountain areas and sparse vegetation. Previously, a 1:50,000 scale stream sediment survey was carried out in the area, and a 1:50,000 scale aeromagnetic study (Yan et al., 2016) delineated multiple anomalies, which are closely related to the copper-nickel ore body. However, these two methods do not have the function of sounding and cannot clearly explore, for example, underground structures or the ore body extension range. Electromagnetic methods (including natural and controlled sources) have been proven to be very useful for imaging the resistivity of subsurface structures (Jones and Garcia, 2003; Gu et al., 2006; Abubakar et al., 2011; Hu et al., 2013).

The magnetotelluric method (MT) is an exploration method that was introduced by the French scholar Cagniard (1953) and former Soviet scholar Tikhonov to study the electrical structure of the underground rock by observing the instantaneous natural electromagnetic field on the ground. The audio frequency magnetotelluric method (AMT) is based on the MT, and the observational device and parameters are largely the same; however, only AMT uses the audio frequency electromagnetic signal. Because the natural field source is distant lightning, there is a signal minimum between 1 kHz and 5 kHz, the so-called "dead band" (Garcia and Jones, 2005, 2008). This weakness in energy results in unreliable data and is more difficult to interpret. In 1975, Goldstein introduced CSAMT using artificial current sources to overcome the limitation of the weak natural signal (Goldstein and Strangway, 1975; Zonge and Hughes, 1991).

Scalar CSAMT transmits electromagnetic field signals from only one fixed grounded dipole. Impedance is defined as the ratio of the orthogonal components Ex, parallel to the source, and Hy, which is perpendicular to the source. For tensor CSAMT, five EM field components (E_x, E_y, H_x, H_y, H_z) are measured for two independent polarizations of the source (i.e., ten fields in total at each site). Impedance elements are obtained at each frequency by solving a series of 2×2 matrix equations determined from the fields due to the two sources (Li and Pedersen,

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1991; Boerner et al., 1993; Wannamaker, 1997). The solutions for the impedance elements in terms of the measured fields are.

$$Z_{xx} = \frac{E_{x1}H_{y2} - E_{x2}H_{y1}}{H_{x1}H_{y2} - H_{x2}H_{y1}} \tag{1}$$

$$Z_{xy} = \frac{E_{x2}H_{x1} - E_{x1}H_{x2}}{H_{x1}H_{y2} - H_{x2}H_{y1}} \tag{2}$$

$$Z_{yx} = \frac{E_{y2}H_{y1} - E_{y1}H_{y2}}{H_{x2}H_{y1} - H_{x1}H_{y2}} \tag{3}$$

$$Z_{yy} = \frac{Ey_1Hx_2 - Ey_2Hx_1}{Hx_2Hy_1 - Hx_1Hy_2} \tag{4}$$

where subscripts 1 and 2 refer to the source bipole number.

Apparent resistivity ρ and impedance phase ϕ can be obtained using the following formulas:

$$\rho_{ij} = \frac{1}{5f}\left|Z_{ij}\right|^2 \tag{5}$$

$$\phi_{ij} = \tan^{-1}\left(\frac{\mathrm{Im}(Z_{ij})}{\mathrm{Re}(Z_{ij})}\right) \tag{6}$$

where subscript i or j refers to x or y, respectively, and f is the frequency.

Tensor CSAMT measures more data, and the imaging is finer relative to the underground electrical structures. However, tensor CSAMT has its own inherent defects, such as the near-field effect, source overprint, and shadow effect, which affect low frequency data and lead to a large reduction in exploration depth.

2. Model calculation and inversion

Fig. 1a shows the model configuration for a large buried conductive prism (10 Ω.m) located within a homogeneous earth (100 Ω.m). The forward calculations of the AMT and tensor CSAMT occur on the survey line. Fig. 1b and Fig. 1c display the 2D inversion results of the AMT data with the "dead band" effect and the tensor CSAMT+AMT data. To visualize the "dead band" effect, the apparent resistivity of the AMT data (at 1000 Hz–5000 Hz) is artificially lowered, and the impedance phase is increased. In Fig. 1b, the shallow part has a low resistance layer and inaccurately reflects the shape of the low resistance body. The red square indicates the projection of the prism. In Fig. 1c, the low resistance part of the inversion is, in general, the same as the red frame in terms of shape or depth.

Thus, we can use the advantages of tensor CSAMT and AMT by applying tensor CSAMT data at high frequency and AMT data at low frequency. This approach can overcome the limitations of the AMT "dead band" effect and the tensor CSAMT near-field effect to increase the exploration depth and improve data quality. Therefore, we conducted a validity test in the Xiarihamu mining area using tensor CSAMT and AMT for known geological profiles.

3. Geological and geophysical characteristics

The mining area, which is in the western East Kunlun Mountains and on the southern rim of the Qaidam Basin, is located in the Qimantag Block. The tectonic unit of this area belongs to the East Kunlun arc basin system of the Qin-Qi-kun orogenic system. This mining area is situated in the Bokalike-xiangride Indosinian Au-Pb-Zn-Cu metallogenic belt that includes the Qimantage-Dulan Variscan Fe-Co-Cu-Pb-Zn-Sn (Sb, Bi) metallogenic belt to the north as well as many large and medium-sized deposits to the west, such as Kendekeke Fe-Co polymetallic deposits, Yemaquan Fe-Zn deposits, and Bayingguole-river iron deposits.

The Jinshuikou group Baishahe Formation and Quaternary strata are the main strata exposed in the Xiarihamu area. The scale of the Baishahe rock formation of the Paleoproterozoic Jinshuikou rock group is small and mainly comprises striped biotite slender gneiss, biotite schist, and marble. The origin of the Quaternary strata, which is distributed extensively in the area and comprises clay, sand and gravel, is more complex and contains, for example, alluvial, proluvial, slope, ice and ice-water piled up, as well as wind-blown deposits.

Electromagnetic measuring lines are arranged in the HS26 anomaly area (Fig. 2). The Cu-Ni-Co ore bodies are distributed in a basic-ultrabasic complex rock mass, and the Paleoproterozoic Baishahe rock formation of the Jinshuikou rock group is the wall rock. The ore bodies are mainly thick strata that are primarily disseminated, with conglomerate ore in the upper part and mostly densely disseminated and tight

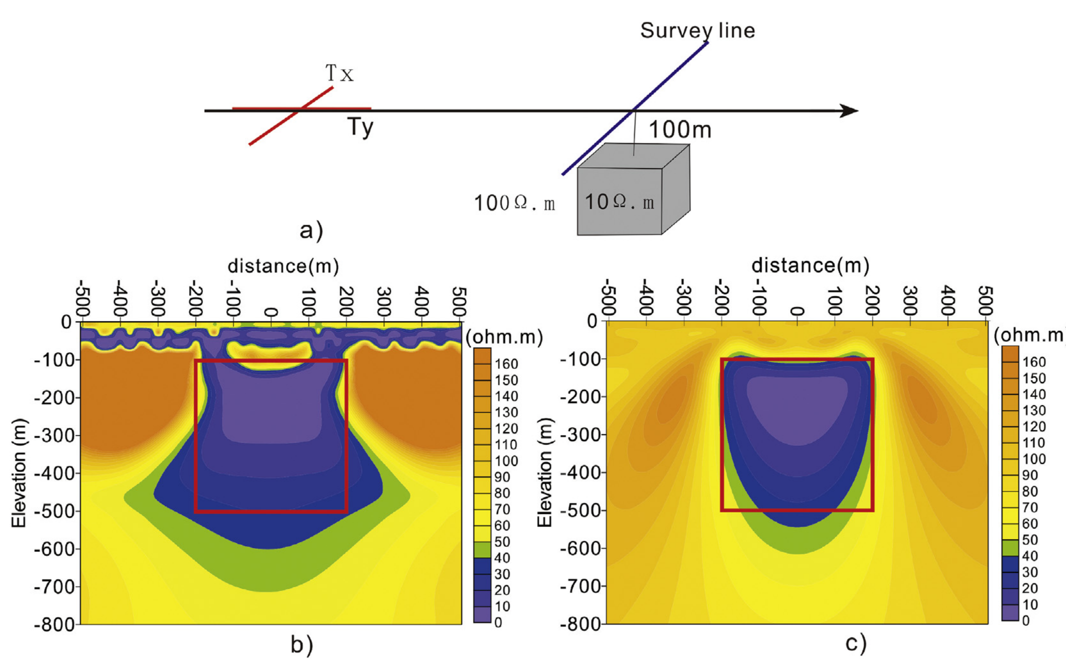

Fig. 1. (a) Model configuration for a buried conductive prism. (b) 2D inversion results of the AMT data with the "dead band" effect. (c) 2D inversion results of the tensor CSAMT+AMT data.

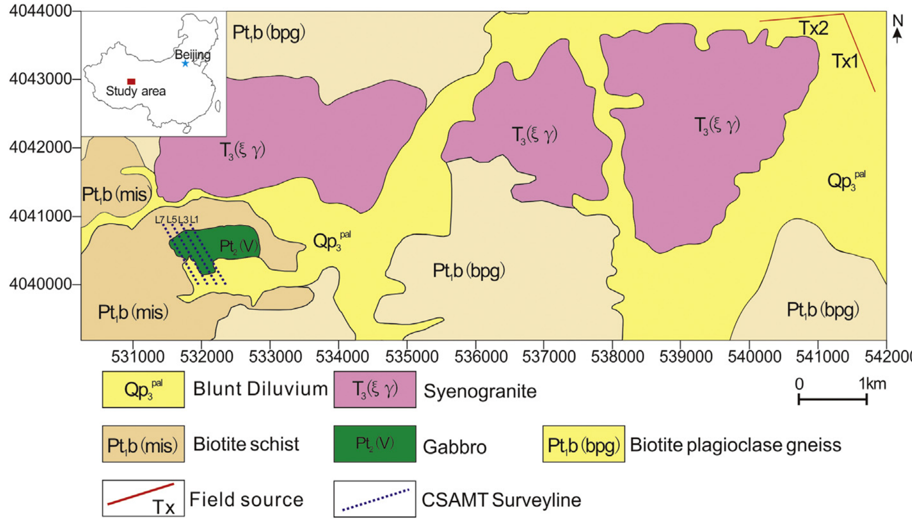

Fig. 2. Simplified geologic map of the Xiarihamu region and the locations of the tensor CSAMT/AMT profiles.

massive ore in the lower and bottom parts. A few lenticular and funnel-like ore bodies are located in the upper part of the rock, having formed top suspensory ore or banded distributions in the rock mass. The ore-bearing rocks are predominantly lherzolite and pyroxenite. The roof of the ore body consists mainly of gabbro, pyroxenite, biotite plagioclase gneiss, and quartz schist. Biotite plagioclase gneiss, granitic gneiss, marble, and quartz schist are present in the base.

The statistical table of resistivity and polarizability of some of the rocks (mines) in this mine area are shown in Table 1(Zhang et al., 2015). Resistivity from 2000 Ω.m to 4000 Ω.m indicates rock mass, such as gabbro, pyroxenite, augite peridotite and wall rocks, such as biotite plagioclase gneiss, granitic gneiss, and quartzite, with a polarizability range of 1%–3.5%. The resistivities of orebodies, such as nickel pyrite pyroxenite and nickel pyrite olivine, are 350 Ω.m-650 Ω.m, with polarizabilities of 20%–40%. There are significant physical differences between the orebodies and wall rocks, and the abnormal characteristics include low resistance and high polarization. Hence, an electromagnetic survey can be carried out in the area.

4. Tensor CSAMT and AMT data acquisition, processing and inversion

4.1. Data acquisition

We arranged 4 lines to cross the major anomaly area in the Xiarihamu deposit (Fig. 2). Many boreholes have been drilled for each line, and the distribution of the lithology and the scope of the ore body are very clear. The distance between lines was 80 m. Two different polarized directional transmitter bipoles (Tx1 and Tx2) for tensor measurements were oriented approximately N21°E and N87°W and were approximately 1.2 km in length each. The distance of the transmitter-receiver was approximately 9.5 km. Tx1 and Tx2 alternately transmitted an electromagnetic field signal to the ground by transmitting current between 4 A (high frequency) and 18 A (low frequency). Tensor CSAMT data were acquired in the effective frequency range of 8 Hz to 8533 Hz using the V8 multifunction receiver and the TXU-30 transmitter (30 kw) manufactured in Phoenix, Canada. The distance between stations was 25 m. The nonpolarized electrode was used to receive the electric field signal, and the inductive magnetic sensor AMTC-30 was used to receive the magnetic field signal. The tensor observation was performed at each station, and two orthogonal electric and three orthogonal magnetic field components were obtained. Following the acquisition of tensor CSAMT data, the controlled source was turned off, and natural source AMT data were obtained. The AMT data were measured using the Phoenix V5 system, Canada. The frequency range of acquisition was 0.35 Hz to 10,400 Hz for each station.

The Xiarihamu copper-nickel deposit, which is located in a depopulated zone, has not yet been mined, and there are few sources of electromagnetic interference in the survey area. The quality of the collected data is reliable, and the signal-to-noise ratio is very high.

Table 1
Resistivity and polarizability means for the main rocks and ores in the survey area.

Rocks or ores	Mean of resistivity (Ω.m)	Mean of polarizability (%)
Biotite plagioclase gneiss	2363	3.20
Quartzite	2104	2.14
Granitic gneiss	2432	1.93
Gabbro	2021	3.39
Pyroxenite	3072	2.25
Augite peridotite	3766	1.66
Nickel pyrite pyroxenite	667	20.88
Nickel pyrite olivine	387	40.22

4.2. Data processing

The tensor CSAMT data observed by the V8 system are the amplitude and phase of the electromagnetic field components. According to the Euler formula, the complex number form of the electric or magnetic field values is.

$$C = A^{*}\left(e^{i\theta}\right) = A^{*}(\cos\theta + i^{*}\sin\theta) \tag{7}$$

where C is the electric or magnetic field, A is the amplitude, θ is the phase, e is the natural constant, and i is the imaginary unit.

Thus, the tensor impedance can be calculated using Formulas (1)–(4), and the apparent resistivity and impedance phases can then be calculated.

The recorded AMT time series data were converted to the frequency domain for each site by discrete Fourier transform (DFT). The calculated DFT values at each frequency were used to create a cross-power spectrum matrix. This matrix can be calculated using standard formulas to get tensor impedance elements, apparent resistivity, and phase, as well as other electrical parameters.

The apparent resistivity and phase of the tensor CSAMT and AMT were plotted against frequency in Fig. 3 for site 27 of line 3. The AMT curve shows the “dead band” effect in the high frequency band (1 kHz–5 kHz), which can cause significant signal amplitude attenuation, while the tensor CSAMT data are good. In the low frequency band

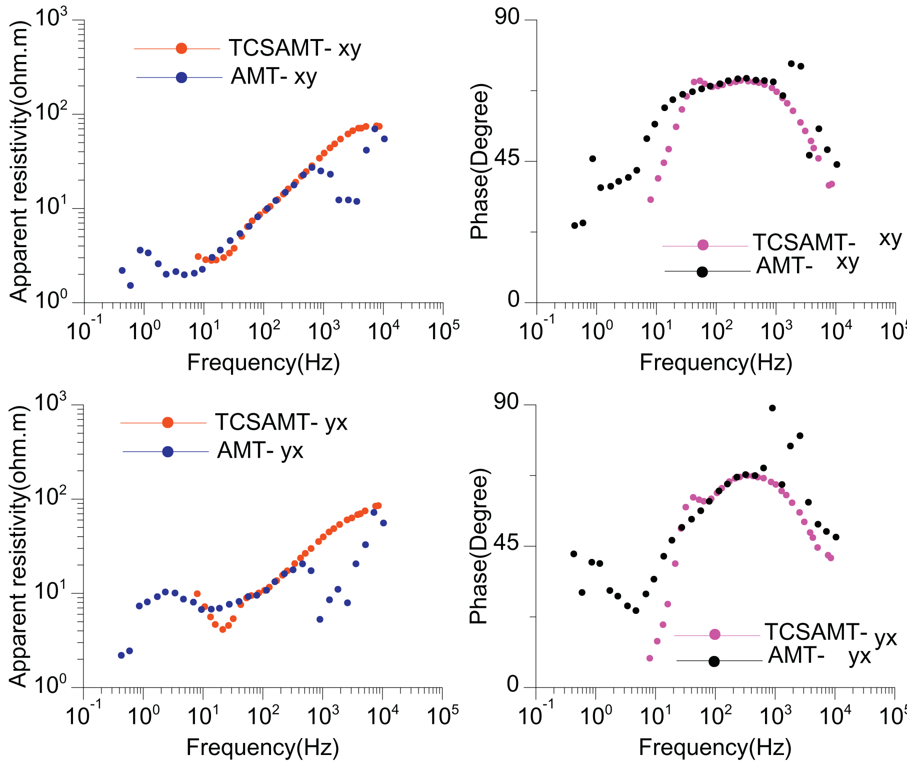

Fig. 3. TCSAMT and AMT data for site 27 of line 3.

(below 100 Hz), the tensor CSAMT data enters the transition zone and near field, and the data below 1 Hz are unstable because the signal is too weak for AMT. Therefore, the splicing tensor CSAMT (200 Hz–8533 Hz) and AMT (1 Hz–200 Hz) data were used for inversion and interpretation.

The static effect occurs if there is a local electrical 2D or 3D heterogeneity near the surface and if the wavelength of the electromagnetic wave is much larger than that of the inhomogeneous body, which will generate an electric field from the accumulated charges and produce galvanic distortion (Jones, 1988). The static effect does not affect the phase curve, only the apparent resistivity curve. The maximum intensity of the impact of up to two orders of magnitude can generate a false steep anomaly and result in an incorrect interpretation. Some of the stations in the Xiarihamu mining area are affected by the static effect. Therefore, we used the spatial domain low-pass filter method (Torres-Verdín and Bostick, 1992) to remove this effect. The Hanning window is an ideal spatial domain low-pass filter with the following formula

$$h(x) = \begin{cases} \left(1 + \cos\dfrac{2\pi x}{W}\right)/W & |x| \le W/2 \\ 0 & |x| > W/2 \end{cases}$$

where W is the window width, and x is the filter length (separation between the sounding points). In practice, $h(x)$ is applied to seven points.

Another problem is that all of the electric fields are affected by uneven topography, which can introduce noise into the data. The measured voltage and the terrain act in opposite directions; in high terrain, the voltage decreases, whereas, in low terrain, the voltage increases. This finding has caused extensive interference with the interpretation of results(Reddig and Jiracek, 1984; Jiracek, 1990) and is known as the terrain effect. With the development of computer technology and numerical simulation techniques, obtaining merely 2D or 3D terrain anomalies and then eliminating the terrain effect is possible (Wannamaker et al., 1986; Chouteau and Bouchard, 1988; Wang et al., 1999; Nam et al., 2006). The inversion software SCS2D used in this study includes the terrain correction function and deals with terrain directly.

4.3. Dimensional analysis

Strictly speaking, although the underground structure is 3D, the closer the underground structure is to 2D, the more realistic the

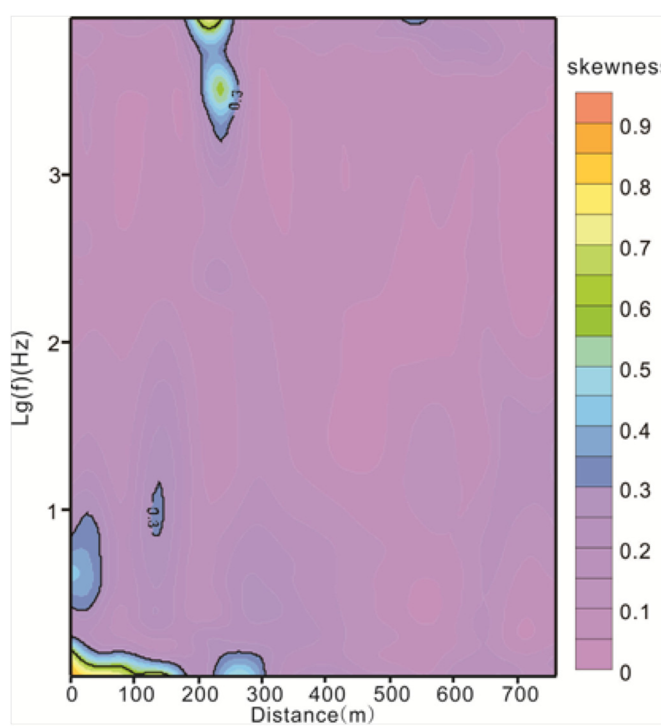

Fig. 4. Contour map of skewness for line 3.

Table 2
Two-dimensional inversion resistivity model meshes and final overall rms misfit for lines 3 and 7.

Model name	Model meshes	Final overall rms misfit
line3	39 × 21	1.68
line7	39 × 21	0.819

geoelectric model obtained by the 2D inversion. We can assess structural dimensionality by the skewness (S), which is defined by the impedance tensor rotation invariant, defined as

$$S = \frac{|Z_{xx} + Z_{yy}|}{|Z_{xy} - Z_{yx}|}$$

which is zero for 2D structures and greater than zero for general 3D structures. The smaller the S, the closer the structure is to 2D, whereas, when S is large, the closer the structure is to 3D. In general, when $S < 0.3$, the electrical structure can be treated approximately as a 2D case (David Beamish, 1986). The contour map of skewness for line 3 is presented in Fig. 4. It is clear that the skewness value of most of the measuring points in the section is below 0.3, with only a few parts >0.3. In general, there are 2D characteristics. The other three lines also show the same dimension; thus, a 2D inversion is reasonable for all the lines in this area.

4.4. 2D inversion

The conventional approaches developed for the MT data can be applied to the joint tensor CSAMT far field data and the AMT low frequency

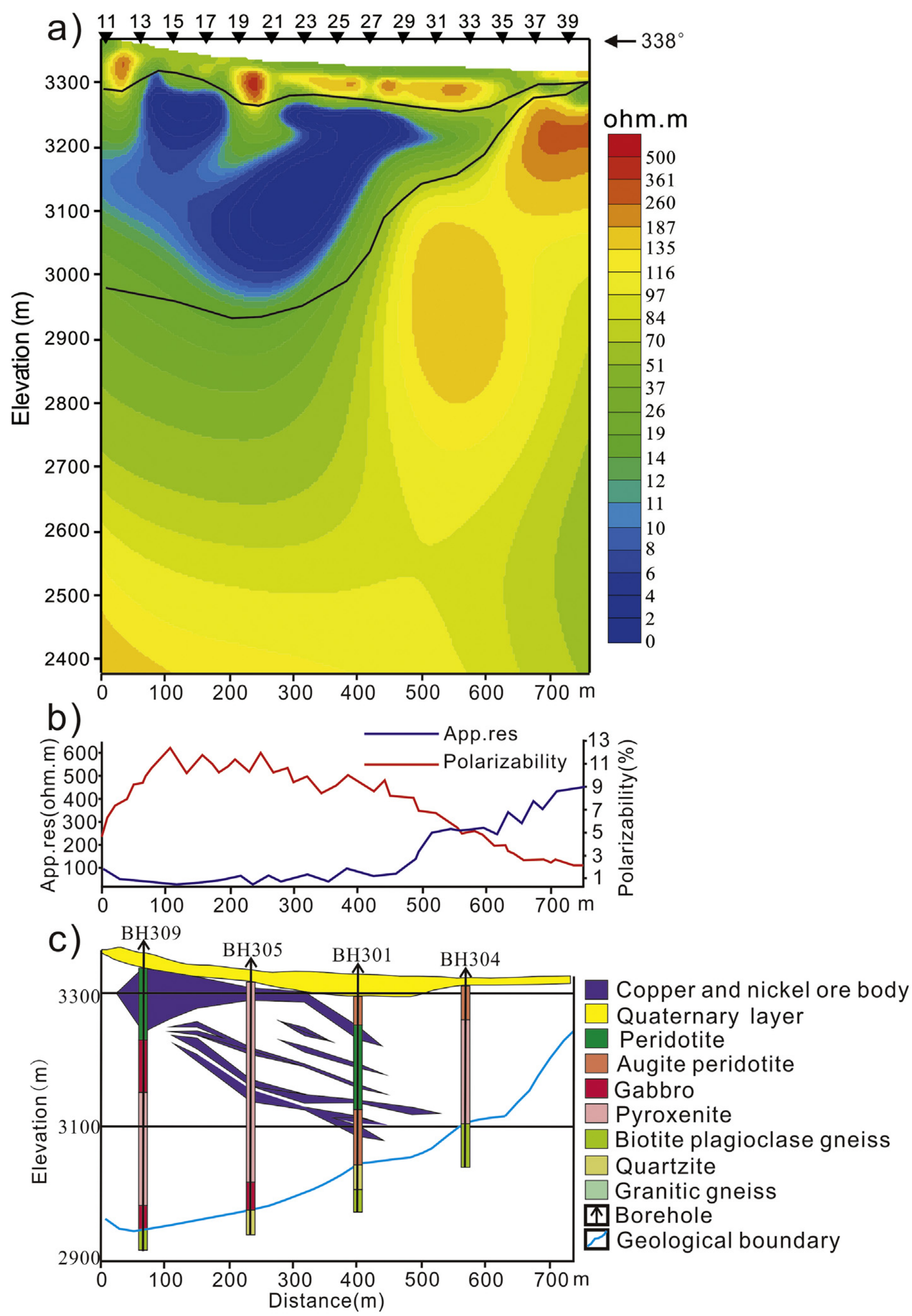

Fig. 5. Integrated profile of line 3. (a) 2D TM + TE mode inversion results of the tensor CSAMT+AMT data. (b) apparent resistivity and polarizability from IP data. (c) sketch of the geologic section.

band data. In this study, a 2D inversion of all lines was carried out using SCS2D software. The use of 2D smooth-model inversion software is a robust method for converting far-field CSAMT or AMT data from a line of soundings to a resistivity model cross-section, which was developed by Zonge International (MacInnes, 1999). This method can invert the apparent resistivity and impedance phase data, whether the data are in TM, TE or TM + TE modes, and finally generates a smooth resistivity model section. The inversion software modeling code uses finite-element approximations to the relevant differential equations (Wannamaker et al., 1987; Goldie, 2002), and the algorithm of inversion is based on smoothness-constrained least-squares methods. The final number of iterations is 5, and the software uses a root-mean-square (rms, observed-calculated data) error criterion to measure data misfit. The misfit between the observed and the calculated data is shown in Table 2.

4.5. Inversion results and integrated interpretation

It is assumed that the direction of the line is the x-direction since ρ_{xy} is the TM polarization, and ρ_{yx} is the TE polarization because the line is arranged in the vertical ore body. The poor-quality data are first removed, and the 2D inversion of the TM, TE and TM + TE modes of the apparent resistivity and impedance phases is performed. The TM + TE mode inversion is the inversion of incorporating the TM and TE mode data. The constraint condition is increased, the accuracy of the inversion result is improved, and the characteristics of the underground geological structure can be imaged more comprehensively. The inversion results of the three modes are analyzed by referring to the geological and geophysical information of the survey area. Finally, the TM + TE mode inversion results are more consistent with the actual geological conditions. For brevity, this paper only presents the results for two of the lines.

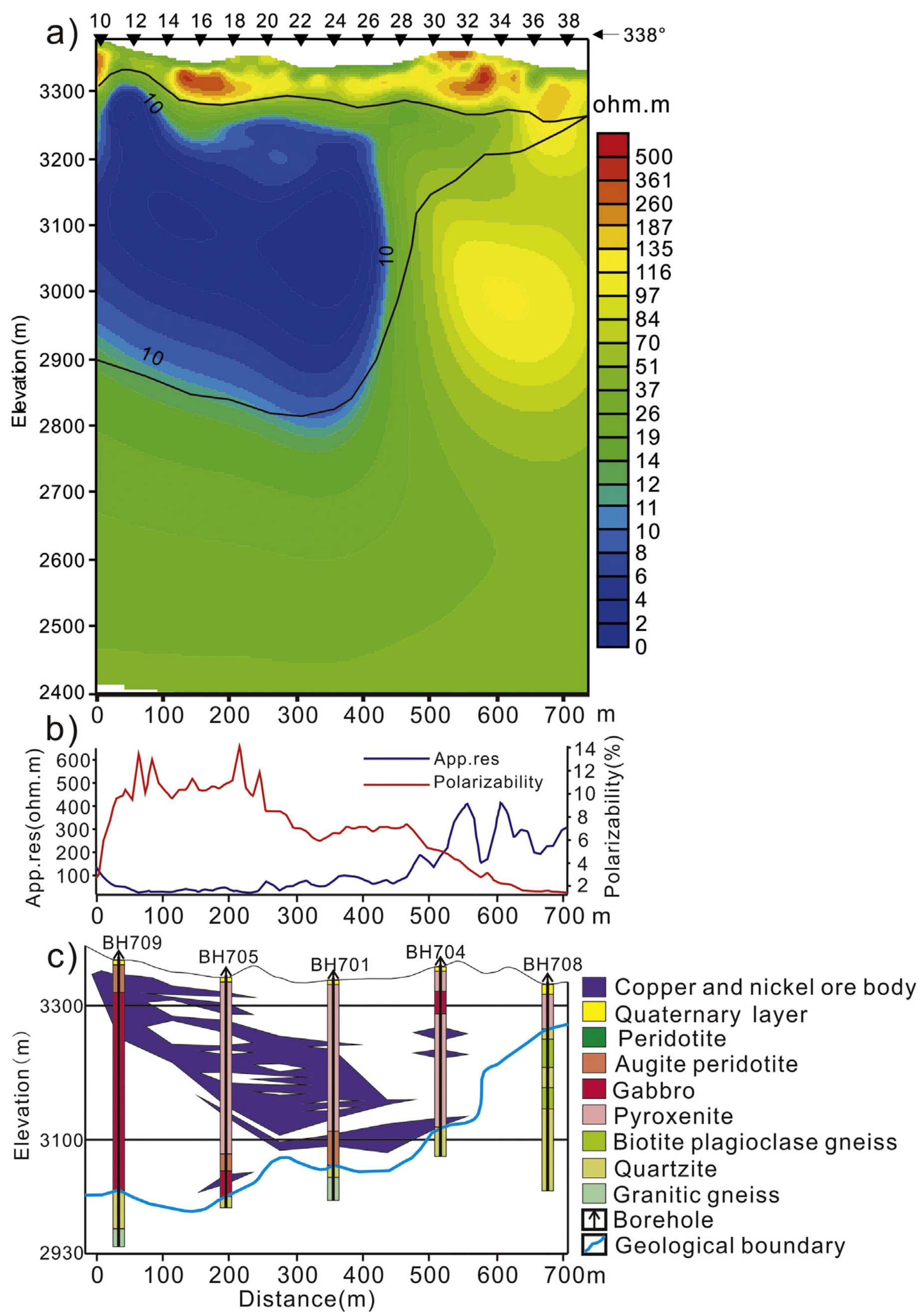

Fig. 6. Integrated profile of line 7. (a) 2D TM + TE mode inversion results of the tensor CSAMT data. (b) apparent resistivity and polarizability from IP data. (c) sketch of the geologic section.

4.6. Profile of line 3

Line 3 is 750 m long; Fig. 5a presents the joint tensor CSAMT and AMT 2D inversion resistivity section. The diagram shows that there are two inferred electrical interfaces (the black solid line). The two electrical interfaces divide the profile into three electrical domains. The first electrical domain (I), which is characterized by medium-high resistivity from 100 to 300 Ω.m, is layered along the terrain, and the thickness of approximately 80 m is presumed to be mixed rock of Quaternary cover and weathered igneous rocks. The second electrical domain (II) is characterized by an extremely low resistivity of <50 Ω.m. This domain is cystic, with a full southward dip, and is thick to the north and thin to the south, with the thickest section from 3300 m to 2950 m in elevation. This characterization corresponds to the low resistance and high polarization anomalies of IP from 50 m to 500 m along the line (Fig. 5b) and matches well with the scope, shape and tendency of the ore body in the geological map (Fig. 5c). The third electrical domain (III), which is characterized by high resistivity from 100 to 500 Ω.m, is the bottom wall rock and belongs to high-resistance metamorphic rock. The electrical interface between domains II and III is, in general, in good agreement with the geological interface on the geological map (Fig. 5c).

4.7. Profile of line 7

Line 7 is also 750 m long and 160 m from line 3. Fig. 6a presents the 2D TM + TE mode inversion results of the tensor CSAMT. The resistivity structure characteristics of this profile are very similar to those of line 3. The two electrical interfaces (black solid line) also divide the section into three domains. However, the bottom interface of the low resistance body in the second domain is at approximately 2800 m in elevation, while the ore body in the geological profile (Fig. 6c) is at approximately 3000 m in elevation, where the low resistance range extends down. This finding occurs because the line does not carry out AMT measurements, and the tensor CSAMT data at approximately 40 Hz in the transition zone and near field, as well as 40 Hz below the frequency data, cannot be used. Therefore, the depth of the inversion is not deep enough. The data are not clear enough to image the bottom interface of the ore body. The low-resistance high-polarization anomalies found by IP from 0 to 500 m along the line (Fig. 6b) are consistent with the ore body range.

5. Conclusions

This research work successfully applied comprehensive geophysical and geological methods to explore the Xiarihamu copper-nickel deposit. We explored 4 lines using tensor CSAMT and AMT. Based on the analysis of the field data, we can conclude the following.

In this mining area, the “low resistance and high polarization” anomaly is caused by the ore body. We applied three methods (tensor CSAMT/AMT/IP) to effectively detect the concealed ore body. The predicted ore bodies are generally consistent with the ore bodies in the geological drilling section.

The IP method can delineate the location of the ore body. This method is suitable for exploring a large area. By using the IP anomaly in the plane, the favorable location of mineralization can be quickly and effectively delineated in the mineralization prospecting area.

Although the tensor CSAMT field acquisition device is complex and relatively inefficient, it can acquire data in two directions at the same measurement point and provide more information than can scalar CSAMT. Therefore, the results of the inversion in complex areas are more accurate than those of the scalar CSAMT.

Combining tensor CSAMT and AMT can overcome the inherent shortcomings of these methods, such as limited exploration depth and the “dead band” effect. In this work, we accurately detected the electrical structure underground, the thickness of the overburden, and the occurrence and the burial depth of the ore body.

In conclusion, the success of this experimental study shows that comprehensive geology and geophysical surveys can effectively detect hidden deposits in the Xiarihamu mining area and provide an effective method and exploration strategy to find similar potential deposits in the surrounding areas.

Acknowledgments

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